

a Biological Model and Silicone Biofilm Substrate Simulated Microgravity and Ground Hardware



Mycobacterium marinum 1218R

BSL-2 surrogate for Mycobacterium tuberculosis

Stable chromosomal RFP at Giles phase subunit RFP3D11, RFP3D39, RFP3D47

1. Lab-Designed 3D Clinostat (*n* = 3)

Continuous 2-axis clinorotation (*I* = 1.5 rpm, *O* = 3.825 rpm)

2. Random Positioning Machine 2.0 (*n* = 3)

Random multi-axis velocity vectoring, time-averaged < 0.01*g*

Samples: RFPRPM4, RFPRPM41, RFPRPM6

3. Static 1*g* Earth Control (*n* = 3)

Stationary incubator shelf adjacent to simulators

Samples: RFPNG14, RFPNG35, RFPNG45

Culture Vessel Specifications (31°C, 4 Days):

- Polydimethylsiloxane (PDMS) silicone membranes
- Suspended biofilm-forming cell pellets harvested
- *n* = 3 biological replicates per experimental modality

c Empirical Multi-Scale Analytical Framework

